Experiment 17

Linear Regression for a given data set

```
import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
df
```

✓ 0.0s

	YearsExperience	Salary
0	1.1	39343
1	1.3	46205
2	1.5	37731
3	2.0	43525
4	2.2	39891
..	...	...
25	9.0	105582
26	9.5	116969
27	9.6	112635
28	10.3	122391
29	10.5	121872

30 rows × 2 columns

```
df.info()
```

✓ 0.0s

```
<class 'pandas.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64  
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

```
df.dropna(inplace=True)
```

✓ 0.0s

```
df.info()
```

✓ 0.0s

```
<class 'pandas.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   YearsExperience  30 non-null     float64
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dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

```
df.describe()
```

✓ 0.0s

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

```
features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values
```

✓ 0.0s

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_stat
    e=23)
```

✓ 0.2s

```
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
```

✓ 0.1s

```
LinearRegression()
```

```
model.score(x_train,y_train)
```

✓ 0.0s

```
0.9603182547438908
```

```
model.score(x_test,y_test)
```

✓ 0.0s

```
0.9184170849214232
```

```
model.coef_
```

✓ 0.0s

```
array([[9281.30847068]])
```

```
model.intercept_
```

✓ 0.0s

```
array([27166.73682891])
```

```
import pickle
pickle.dump(model,open('SalaryPred.model','wb'))
```

✓ 0.0s

```
model=pickle.load(open('SalaryPred.model','rb'))
```

✓ 0.0s

```
yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)
```

✓ 0.0s

```
Enter Years of Experience: 44
```

```
print("Estimated Salary for {} years of experience is {}: " .format(yr_of_exp,Salary))
```

✓ 0.0s

```
Estimated Salary for 44.0 years of experience is [[435544.30953887]]:
```